

Homework 18: One-to-One & Onto Linear Transformations

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BCS A, E, F, G

Definition 1. One-to-One & Onto Transformation. A linear transformation $T: V \rightarrow W$ is called *one-to-one* if T maps distinct vectors in V to distinct vectors in W . If $\text{range}(T) = W$, then T is called *onto*.

Remark 1. The definition of one-to-one may be written more formally as follows:

$T: V \rightarrow W$ is one-to-one if, for all $\mathbf{u}$ and $\mathbf{v}$ in V , $\mathbf{u} \neq \mathbf{v}$ implies that $T(\mathbf{u}) \neq T(\mathbf{v})$.

A more familiar perspective from single variable calculus which can be used conveniently is an equivalent statement to the one above is:

$T: V \rightarrow W$ is one-to-one if, for all $\mathbf{u}$ and $\mathbf{v}$ in V , $T(\mathbf{u}) = T(\mathbf{v})$ implies that $\mathbf{u} = \mathbf{v}$.

Onto Transformation. Another way to write the definition of onto is as follows:

$T: V \rightarrow W$ is onto if, for all $\mathbf{w}$ in W , there is at least one $\mathbf{v}$ in V such that $\mathbf{w} = T(\mathbf{v})$.

In other words, given $\mathbf{w}$ in W , does there exist some $\mathbf{v}$ in V such that $\mathbf{w} = T(\mathbf{v})$? If, for an arbitrary $\mathbf{w}$, we can solve this equation for $\mathbf{v}$, then T is onto.

Problem 1. Which of the following linear transformations are one-to-one? onto?

a) $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ x - y \\ 0 \end{bmatrix}$.

b) $D: P_3 \rightarrow P_2$ defined by $D(p(x)) = p'(x)$.

c) $T: M_{22} \rightarrow M_{22}$ defined by $T(A) = A^T$.

Theorem 1. A linear transformation $T: V \rightarrow W$ is one-to-one if and only if $\ker(T) = \{\mathbf{0}\}$.

Problem 2. Show that the linear transformation $T: \mathbb{R}^2 \rightarrow P_1$ defined by $T \begin{bmatrix} a \\ b \end{bmatrix} = a + (a + b)x$ is one-to-one and onto.

Theorem 2. Let $\dim V = \dim W = n$. Then a linear transformation $T: V \rightarrow W$ is one-to-one if and only if it is onto.

Theorem 3. A linear transformation $T: V \rightarrow W$ is invertible if and only if it is one-to-one and onto.

Problem 3. Show that $T: \mathbb{R}^2 \rightarrow P_1$ defined in problem 2 is invertible.

Problem 4. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear operator defined by the formula

$$T(x_1, x_2, x_3) = (3x_1 + x_2, -2x_1 - 4x_2 + 3x_3, 5x_1 + 4x_2 - 2x_3).$$

Determine whether T is one-to-one; if so, find $T^{-1}(x_1, x_2, x_3)$.

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Good Luck